THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Magnetic field lines never intersect. Reason (R): The field at a point has only one direction. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A straight wire with current is placed between the poles of a magnet. The force on the wire is greatest when the wire makes an angle with the field of 1

(a) 0 degrees **(b)** 45 degrees **(c)** 90 degrees **(d)** 30 degrees

3. A generator coil of frequency 50 Hz is rotated at double the speed. The frequency becomes 1

(a) 25 Hz **(b)** 50 Hz **(c)** 200 Hz **(d)** 100 Hz

4. Assertion (A): A straight current-carrying wire produces a magnetic field around it. Reason (R): A compass needle placed near the wire gets deflected when the current flows. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A coil moves along the field lines of a uniform field in a direction parallel to the field. The induced current is 1

(a) alternating **(b)** maximum **(c)** half **(d)** zero

6. A magnet is brought toward a coil at a speed v and the induced emf is E. If the speed is tripled, the emf becomes 1

(a) E **(b)** 9 E **(c)** E / 3 **(d)** 3 E

7. Assertion (A): Soft iron is used as the core of an electromagnet. Reason (R): Soft iron is easily magnetised and demagnetised. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. A bar magnet is brought towards a closed coil of wire. The coil 1

(a) never develops a current
(b) develops an induced current while the magnet moves
(c) develops a current even if the magnet is at rest
(d) becomes a permanent magnet

9. **Assertion (A):** Two magnetic field lines never intersect each other. 1
Reason (R): At any point in a magnetic field, the field has only one direction.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A solenoid carrying current behaves like a bar magnet. 1
Reason (R): The magnetic field inside a long solenoid is uniform.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A charged particle enters a magnetic field and its speed remains unchanged. Explain why. 2

12. Calculate the current drawn by a 2.2 kW geyser at 220 V and say whether a 5 A or a 15 A circuit is suitable. 2

13. Why is a switch always connected to the live wire? 2

14. The frequency of AC changes from 50 Hz to 60 Hz. Find the new period and compare the number of reversals per second. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the force on a current-carrying conductor depends on the angle between the conductor and the field. 3

16. Why should an electrical appliance such as a toaster have a metal casing connected to the earth wire? 3

17. Why do we say that electricity and magnetism are related? Give two examples. 3

OR

A wire carrying current is bent in the form of a circular loop. How are the field lines inside the loop and outside it different? 3

18. Explain the difference between the nature of the magnetic field of a straight wire and a bar magnet. 3

OR

How would you make a stronger electromagnet using the same wire and battery? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Rohit placed a compass needle near a straight copper wire connected to a battery through a switch. When the switch was closed, the needle was deflected, and when he reversed the battery connections, the needle was deflected in the opposite direction. 5
(i) Name the scientist who first observed this effect.
(ii) What does this experiment show?
(iii) Name the rule used to find the direction of the magnetic field around a straight current-carrying wire.
(iv) If the current in a vertical wire flows upwards, what is the direction of the field lines when seen from above, and how does the field strength change as one moves away from the wire?

20. Describe the construction and working of a simple electric motor, giving the role of the split-ring commutator. 5

OR

In an experiment a bar magnet is pushed into a coil with 3 different speeds, and the deflections of the galvanometer are 2, 4 and 6 divisions. What do you conclude? Also predict the deflection if the coil had double the turns. 5

END OF QUESTION PAPER — Total: 40 Marks